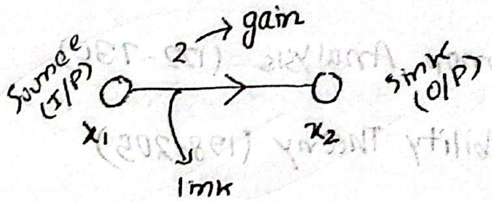
11-07-2026

Signal Flow Graph (SFG):

Block Reduction	Vs SFG
1. Complex Approach	1. Easy Approach
2. Time consuming	2. Less Time Consuming
3. Iterative Process	3. Simple Process / Operation
4. Works with feedback Block	4. Works with feedback loop

Terminology:

① Node : Simple Connector or dot

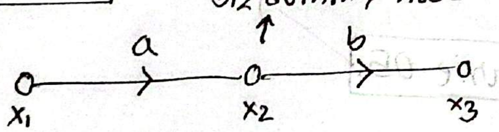


(i) $I/P = x_1$

$O/P = 2x_1 = x_2$

এর কোনো কাজ নেই, তাই dummy node.

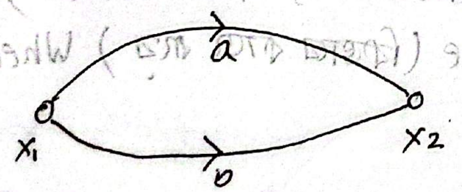
(ii)



$x_2 = ax_1$

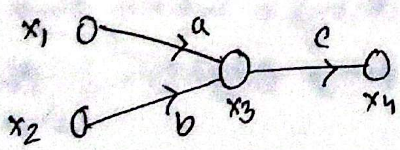
$x_3 = bx_2 = bax_1$

(iii)



$\boxed{x_2 = ax_1 + bx_1 = (a+b)x_1}$

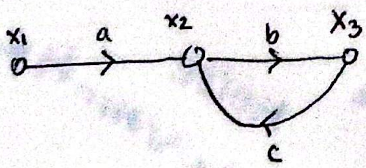
(iv)



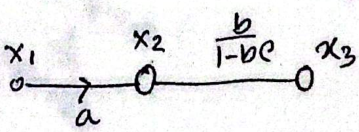
$x_3 = ax_1 + bx_2$, $x_4 = cx_3$

$\therefore x_4 = acx_1 + bcx_2$

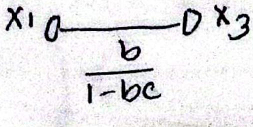
(v)



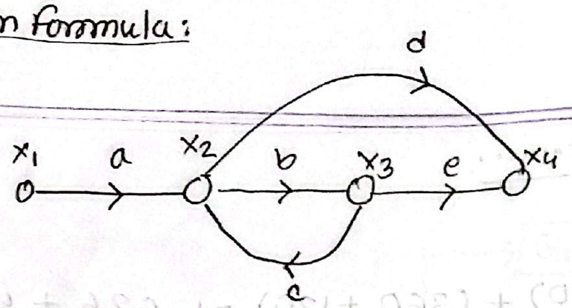
$\frac{b}{1-bc}$ So,



$x_3 = \frac{ab}{1-bc} x_1$



Mason Formula:



→ forward

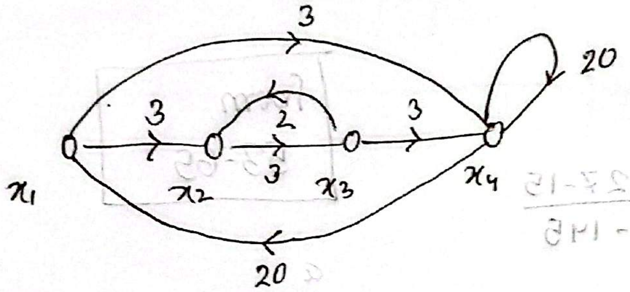
what is path:

$g_1 = x_1 \to x_2 \to x_3 \to x_4$

$\therefore g_1 = abe$

$g_2 = x_1 \to x_2 \to x_4$

$\therefore g_2 = ad$



Soln:

① Path:

$g_1 = 27$

$g_2 = 3$

② ~~Non touching~~ loop:

$L_1 = 3 \times 20 = 60$

$L_2 = 6$

$L_3 = 540$

$L_4 = 20$

③ Non-touching Loop:

L_1 touch করছে না L_2 কে

$L_1' = L_1 \times L_2 = 60 \times 6 = 360$

$L_2' = L_2 \times L_4 = \cancel{60} \times 20 = 120$

Masoon Gain formula: use to find the transfer function

Step by step:

① Draw Signal flow graph

(ii) Forward Path

(iii) Loop findout

(iv) Non- touching loop findout

$$\Delta = 1 - \Sigma \text{ Single loops} + \Sigma \text{ two non touching loops} - \Sigma \text{ three non touching loops}$$

$\Delta_1, \Delta_2, \cdots$ (যতটা Path ততটা আর Δ আলাদা)

$\Delta_1 = 1 - \Sigma$ Single loops ~~not~~ touching path g_1

$\Delta_2 = 1 - \Sigma$ Single loops Not Touching Path g_2

Transfer function $= \dfrac{g_1\Delta_1 + g_2\Delta_2 + \cdots}{\Delta}$

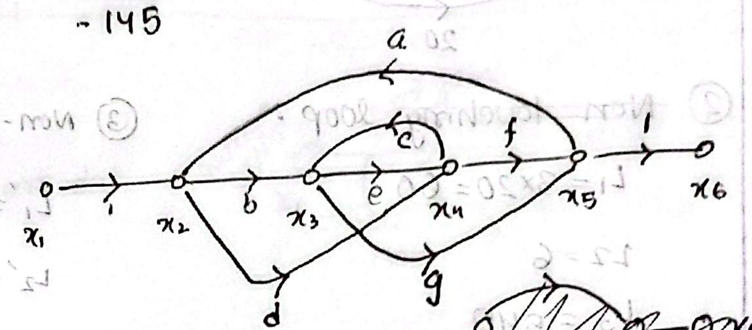
Now, $\Delta = 1-(60+6+540+20)+(360+120) = 1-626+480 = -145$

$\Delta_1 = 1-0 = 1$

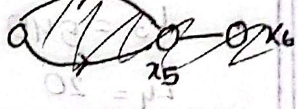
$\Delta_2 = 1-6 = -5$

from 53-65

$\therefore \text{TF} = \dfrac{\Delta_1 g_1 + \Delta_2 g_2}{\Delta} = \dfrac{27-15}{-145}$



$\boxed{1.40}$ Line Equation



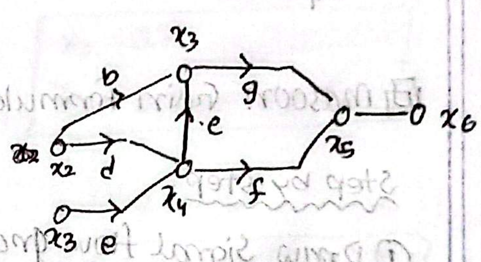
$x_2 = x_1 + a x_5$

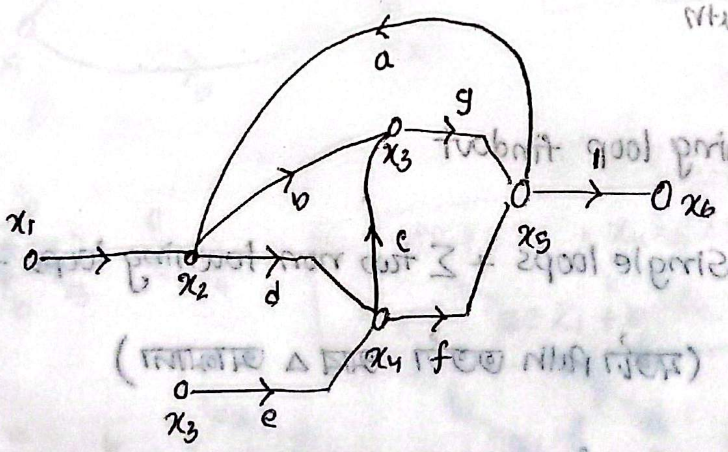
$x_3 = b x_2 + c x_4$

$x_4 = d x_2 + e x_3$

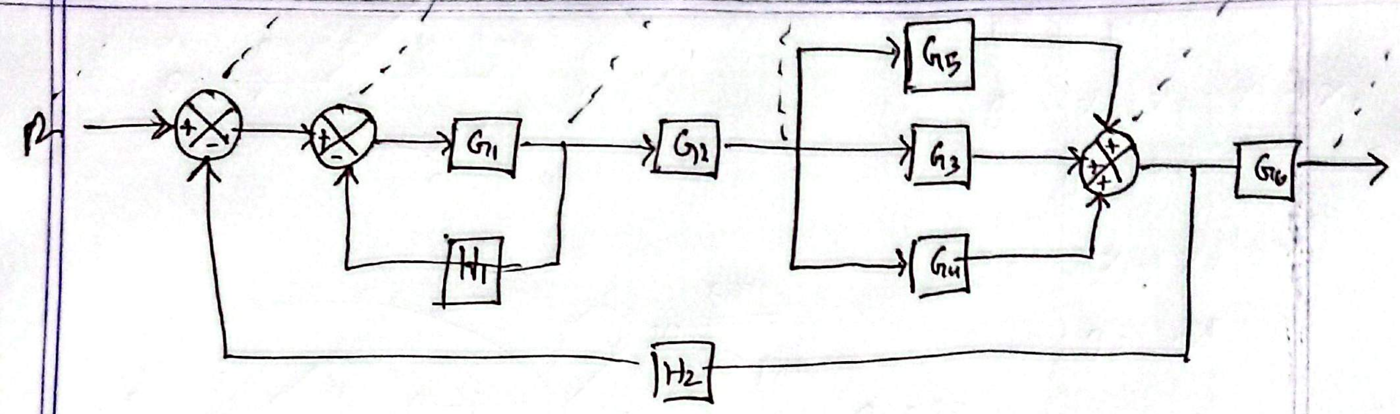
$x_5 = f x_4 + g x_3$

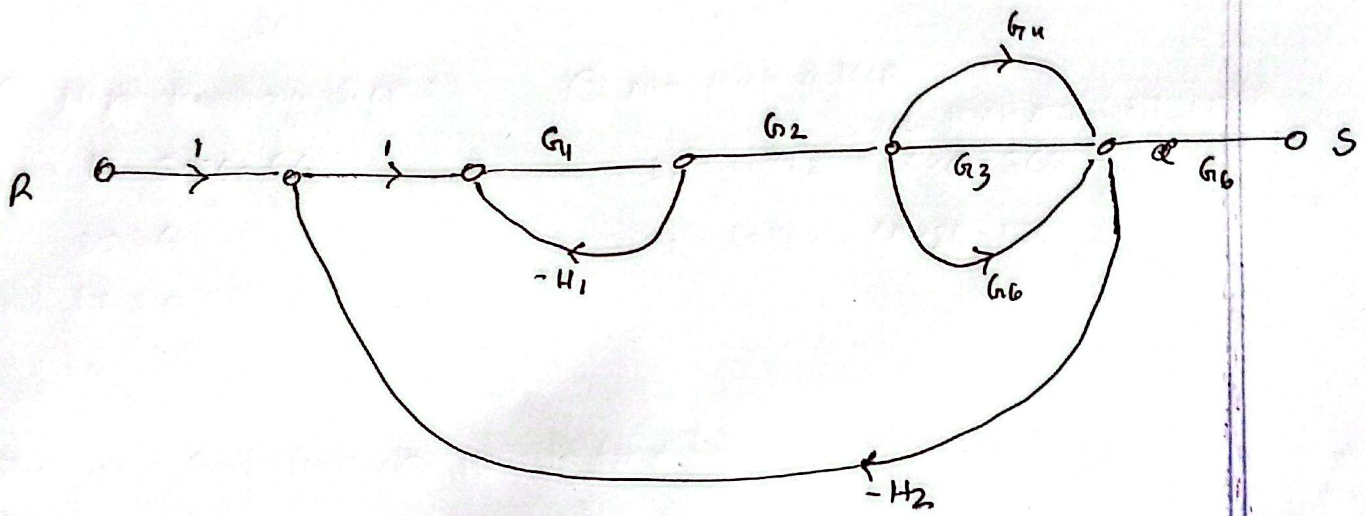
$x_6 = x_5$



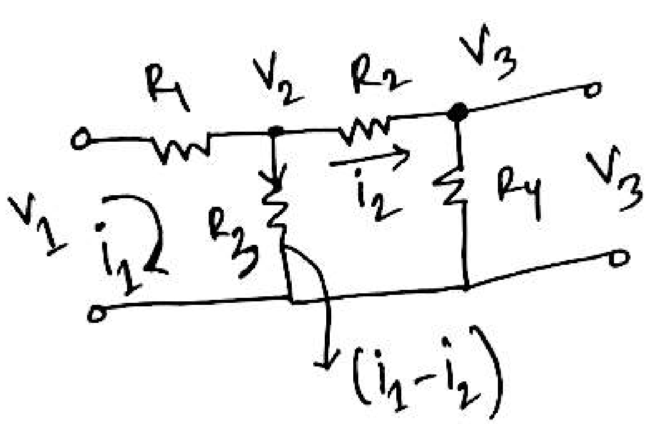


1.43, 42, 41-46





1.47)



$$i_1 = \frac{V_1 - V_2}{R_1}$$

$$V_2 = (i_1 - i_2) R_3$$

$$i_2 = \frac{V_1 - V_3}{R_2}$$

$$V_3 = i_2 R_4$$